LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Quick Revision: Life Processes

Nutrition

- **Autotrophic** (plants): photosynthesis. $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$ (sunlight, chlorophyll)
- Events: (1) light absorbed by chlorophyll, (2) light energy → chemical energy, water split into H and O, (3) CO_2 reduced to carbohydrate.
- Stomata allow gas exchange. Guard cells open and close them (they swell when water flows in).
- **Heterotrophic:** holozoic (Amoeba, humans), parasitic (Cuscuta, ticks), saprophytic (fungi).

Part	Secretion	What it does
Mouth	Saliva (salivary amylase)	Starch → sugar
Stomach	HCl, pepsin, mucus	Acidic medium; protein digestion; protects the lining
Liver	Bile	Makes the medium alkaline; emulsifies fats
Pancreas	Trypsin, lipase	Proteins; emulsified fats
Small intestine	Intestinal juice	Final digestion: proteins → amino acids, carbohydrates → glucose, fats → fatty acids + glycerol. Villi absorb food
Large intestine	n/a	Absorbs water; waste removed through the anus

Respiration

- Glucose → pyruvate (in the **cytoplasm**). Then:
- No oxygen (yeast): pyruvate → ethanol + CO_2 + energy (fermentation).
- Lack of oxygen (muscles): pyruvate → lactic acid + energy (causes **cramps**).
- Oxygen present (**mitochondria**): pyruvate → $CO_2 + H_2O$ + much more energy.
- Energy is stored as **ATP**. Alveoli give a huge surface area with thin walls and capillaries. Haemoglobin carries O_2.

Transportation

- Heart: 4 chambers. Right side carries deoxygenated blood, left side oxygenated. Ventricles have thicker walls. Valves stop backflow.
- **Double circulation:** blood passes through the heart twice per cycle, keeping oxygenated and deoxygenated blood separate. Fish have a 2-chambered heart.
- Arteries carry blood away from the heart (thick, elastic walls). Veins carry blood to the heart (have valves). Capillaries are one cell thick.
- Platelets help clotting. Lymph carries digested fat and drains extra fluid.
- Plants: **xylem** carries water and minerals (root pressure + transpiration pull). **Phloem** carries food (translocation, uses ATP).

Excretion

- Kidneys → ureters → urinary bladder → urethra. Each kidney has many **nephrons**.
- Nephron: glomerulus in Bowman's capsule filters blood → tubule reabsorbs glucose, amino acids, salts and water → urine.
- **Dialysis** (artificial kidney) filters blood but has no reabsorption.
- Plants: excess water by transpiration; wastes stored in old leaves, bark, or as gums and resins.

Quick check

1. Where in the cell does pyruvate break down completely in aerobic respiration?

2. Why do ventricles have thicker walls than atria?

3. Name the tissue that transports food in plants.